Project Update
A Closed-Form QUBO for Graph Partitioning and Image Segmentation

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Abstract

We formulate minimum vertex cover and Boykov–Jolly seeded image segmentation as a *single* quadratic unconstrained binary optimization (QUBO) objective and solve both with classical heuristics, validating every result against an exact reference (exhaustive search; maximum flow). The codebase is complete and tested across three phases (vertex cover, segmentation, and a graph-reconstruction "bridge"), with synthetic, real (32×32 Weizmann horses), and high-resolution (128×128) experiments. This update reports a new *classical solver portfolio* that benchmarks simulated annealing, tabu search, and steepest descent on the identical QUBO, and lays out a no-quantum roadmap for increasing the project's research value.

1 Status

The project encodes two superficially different problems in one energy

$$H(x) = \sum_i a_i x_i + \sum_{i<j} b_{ij}\, x_i x_j, \qquad x_i \in \{0,1\}, \tag{1}$$

and minimizes it with one heuristic sampler. Three phases are implemented and unit-tested (22 tests passing; `ruff`+`black` clean):

- **Phase 1 – Minimum vertex cover.** $H(x) = \sum_i x_i + P \sum_{(u,v)\in E}(1-x_u)(1-x_v)$, $P > 1$; exact check by exhaustive search.
- **Phase 2 – Seeded segmentation (Boykov–Jolly).** $E(x) = \sum_i D_i(x_i) + \sum_{(i,j)} w_{ij}(x_i - x_j)^2$; submodular, so the exact optimum is a maximum-flow min-cut.
- **Phase 3 – Graph reconstruction (GraphMI bridge).** The same smoothness objective, over edge-indicator variables, recovers a hidden graph from released node features; checked against an analytic optimum.

Headline results so far. Table 1 summarizes validation against the exact references.

An integrity note. The first real-image run produced *negative* optimality gaps (the heuristic appearing to beat the "exact" optimum), which is impossible for a submodular energy. The cause was numerical: a large seed penalty next to vanishingly small smoothness weights gave the floating-point max-flow a $\sim 10^{16}$ dynamic range, on which the solver returned a non-minimum cut. Encoding seeds as *bounded* infinities and using integer-scaled capacities fixed it; gaps are now non-negative everywhere. An exact reference is only as trustworthy as its numerics.

Table 1: Validation status. "Gap" is the energy difference to the exact optimum; IoU is against ground-truth masks where available.

Phase	Instances	Result
Vertex cover	12 graphs (≤ 20 nodes)	12/12 valid covers; **12/12 reach the exact optimum**
Segmentation (synthetic)	blobs/squares 16×16	gap = 0, success rate 1.00
Segmentation (real horses)	16 × 32×32 + GT	11/16 optimal; mean IoU 0.45 (up to 0.76)
Segmentation (high-res)	cameraman/coins/cell, 128×128	IoU 0.88–1.00 on clean objects
Graph reconstruction	cycle/path/ER/Petersen	mean edge-F1 0.64 (up to 0.88)

2 New this update: a classical solver portfolio

Following the methodology of Q-Seg [5] (which benchmarks against a classical solver), we now run *several* classical heuristics on the identical QUBO and score each by its optimality gap to the exact reference and its wall-clock time (Table 2). Simulated annealing (SA), tabu search, and steepest descent all consume the same `dimod` model.

Table 2: Classical solver portfolio on the identical QUBO (100 reads). SA and tabu reach the exact optimum on both problem classes; steepest descent ("Greedy") solves vertex cover but **fails on segmentation**, stalling in local minima of the rugged labeling landscape.

Problem	Solver	Mean gap	Max gap	Mean time (s)	Success
Vertex cover	SA	0.000	0.000	0.013	1.00
Vertex cover	Tabu	0.000	0.000	2.100	1.00
Vertex cover	Greedy	0.000	0.000	0.000	1.00
Segmentation	SA	0.000	0.000	0.083	1.00
Segmentation	Tabu	0.000	0.000	2.101	1.00
Segmentation	Greedy	13.20	21.56	0.002	0.00

The takeaway sharpens the project's message: the *formulation* is solver-agnostic, but the *difficulty* is not. A trivial greedy descent suffices for vertex cover yet is useless for segmentation, whereas SA and tabu solve both – a concrete, measured statement about which problems the QUBO move makes easy or hard.

3 Roadmap to increase effectiveness (no quantum hardware)

The project's unique value is *exactly validated, hand-designed objectives*, not state-of-the-art segmentation. The most valuable improvements therefore sharpen the solver comparison, add problems that retain an exact check, and deepen the privacy bridge.

Tier 1 (keeps exact validation). (i) Extend the solver portfolio with parallel tempering and an exact MIP solver (Gurobi/OR-Tools) and report time-to-solution. (ii) Add a *binary image denoising* phase (Greig et al., 1989 [4]): an Ising/MRF energy whose exact optimum is again a maximum-flow min-cut – reinforcing the "one objective, many problems" thesis nearly for free. (iii) GrabCut-style iterated graph cut to raise real-image IoU while staying hand-designed and exact per iteration. (iv) Statistical rigor: ≥ 10 seeds, 95% confidence intervals, time-to-target.

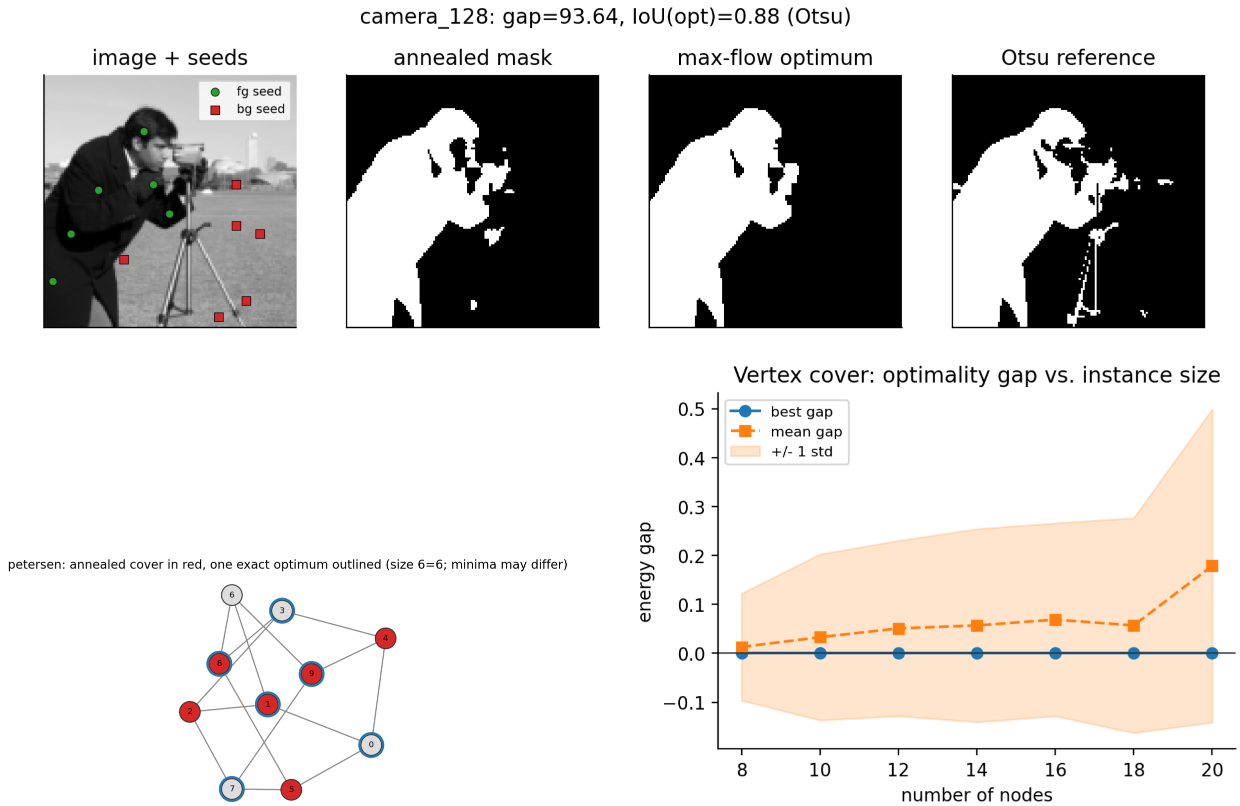

Figure 1: Top: high-resolution segmentation – the annealed mask is identical to the exact maximum-flow optimum and cleaner than an Otsu threshold. Bottom left: a vertex-cover instance (annealed cover in red, one exact optimum outlined). Bottom right: the optimality gap grows gently with instance size while the best read stays optimal.

Tier 2 (research upside). Deepen the GraphMI bridge – reconstruct from a small trained GNN, and plot recovery F1 against feature noise and graph density (a privacy/utility curve), directly serving the lab's work on graph-data privacy. Add a Max-Cut / community-detection QUBO to fill out the "graph partitioning" half of the title.

Tier 3 (scope expansion, trades the exact check). Multi-label segmentation via α-expansion; note that multiway cut is not submodular, so the clean max-flow reference is lost for > 2 labels.

4 Plan, reproducibility, and scope

A day-by-day two-week schedule to a submittable final (report, poster, slides, reproducibility harness, CI) is recorded in `docs/SCHEDULE.md`. The code is public, tested, linted, and runs end-to-end on a laptop or via the included Colab notebook; nothing requires quantum hardware. **Scope:** nothing is learned from data; inputs are small and fixed-size so the exact reference stays computable and the optimality gap remains the primary, interpretable measure of success.

References

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[3] Y. Boykov, V. Kolmogorov. *An experimental comparison of min-cut/max-flow algorithms.* IEEE TPAMI, 2004.

[4] D. Greig, B. Porteous, A. Seheult. *Exact maximum a posteriori estimation for binary images.* JRSS-B, 1989.

[5] S. M. Venkatesh et al. *Q-Seg: Quantum annealing-based unsupervised image segmentation.* arXiv:2311.12912, 2024.

[6] Z. Zhang et al. *GraphMI: Extracting private graph data from graph neural networks.* IJCAI, 2021.